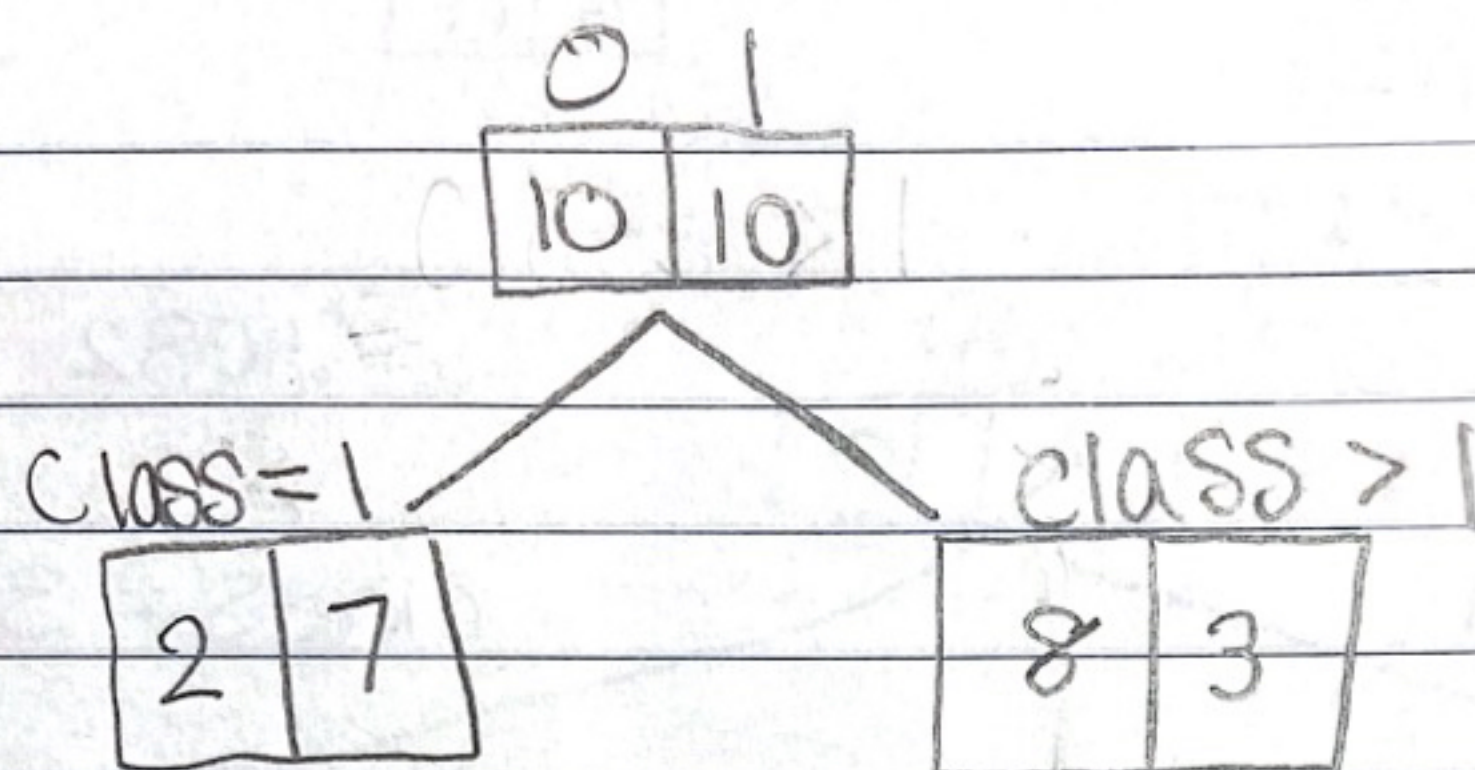


①

Split 1 Options: $I_p = 1 - \left(\frac{10}{20}\right)^2 - \left(\frac{10}{20}\right)^2 = .5$

1.



$$P_0 = \frac{2}{9} \quad P_1 = \frac{7}{9}$$

$$I_L = 1 - \left(\frac{2}{9}\right)^2 - \left(\frac{7}{9}\right)^2 = .3457$$

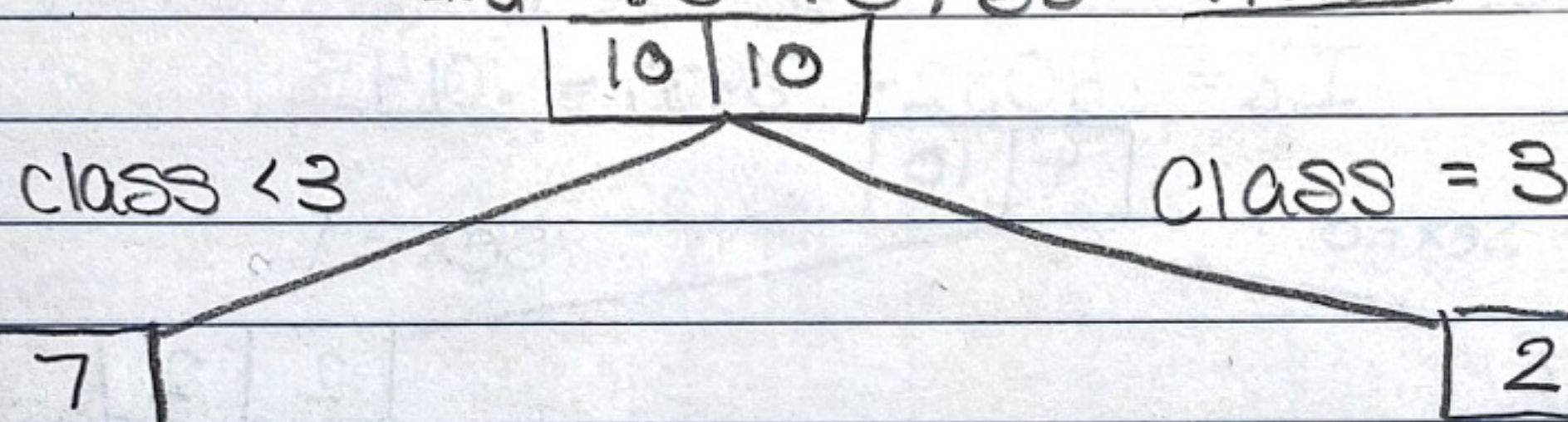
$$P_0 = \frac{8}{11} \quad P_1 = \frac{3}{11}$$

$$I_R = 1 - \left(\frac{8}{11}\right)^2 - \left(\frac{3}{11}\right)^2 = .3967$$

$$I_c = \frac{9}{20}(.3457) + \left(\frac{11}{20}\right)(.3967) = .3738$$

$$I_G = .5 - .3738 = .1262$$

2.



$$P_0 = \frac{8}{15} \quad P_1 = \frac{7}{15}$$

$$I_L = 1 - \left(\frac{8}{15}\right)^2 - \left(\frac{7}{15}\right)^2 = .4978$$

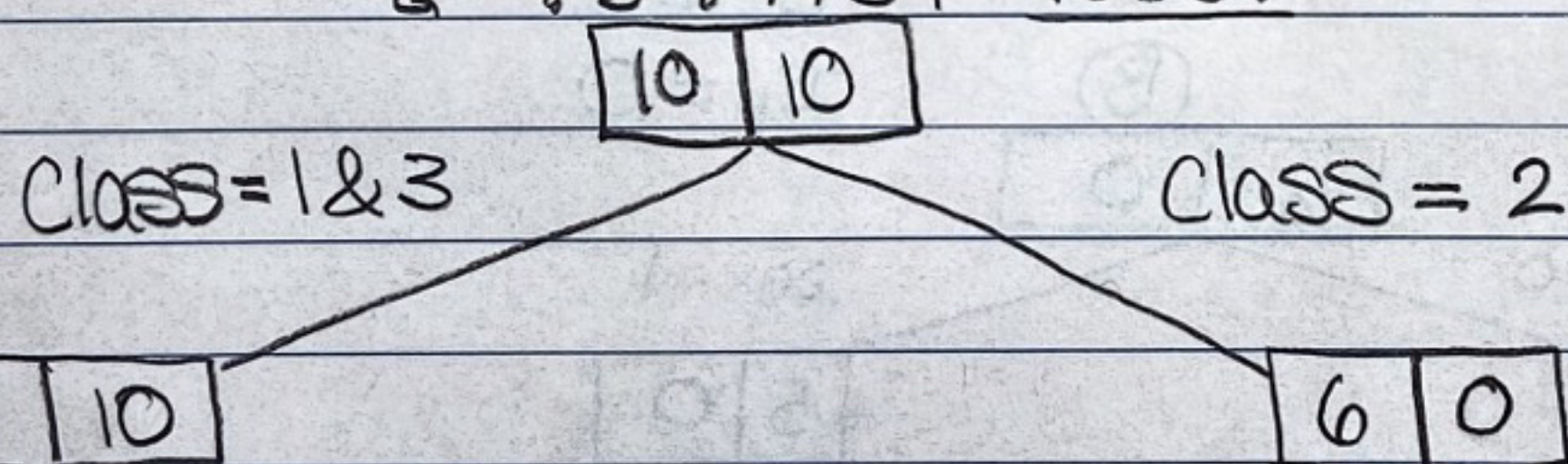
$$P_0 = \frac{2}{5} \quad P_1 = \frac{3}{5}$$

$$I_R = 1 - \left(\frac{2}{5}\right)^2 - \left(\frac{3}{5}\right)^2 = .48$$

$$I_c = \frac{15}{20}(.4978) + \frac{5}{20}(.48) = .4934$$

$$I_G = .5 - .4934 = .0066$$

3.



$$P_0 = \frac{4}{14} \quad P_1 = \frac{10}{14}$$

$$I_L = 1 - \left(\frac{4}{14}\right)^2 - \left(\frac{10}{14}\right)^2 = .4082$$

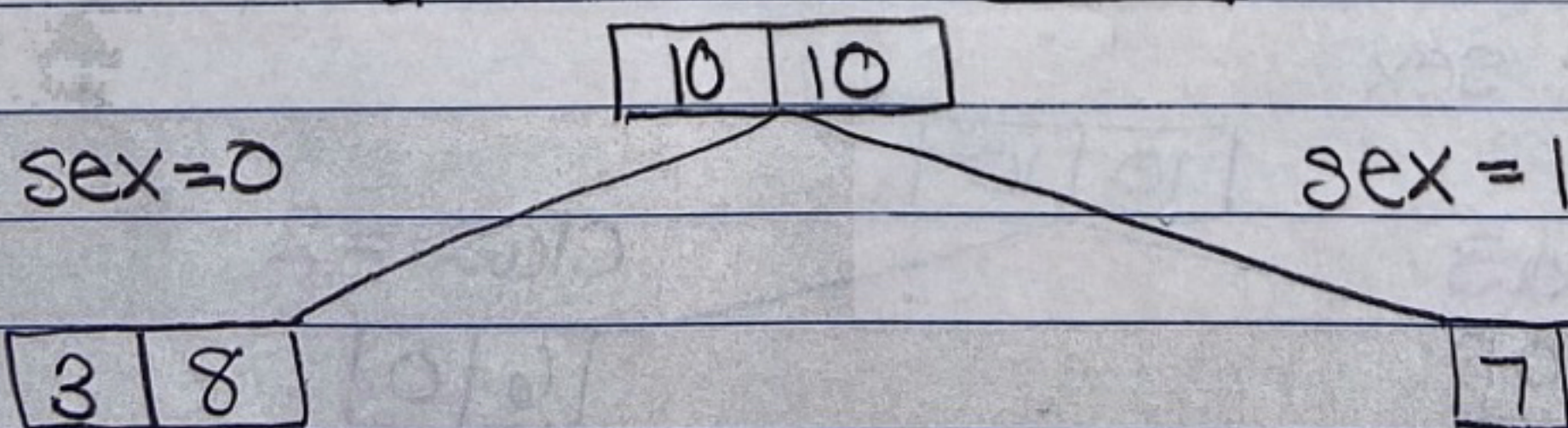
$$P_0 = 1 \quad P_1 = 0$$

$$I_R = 1 - 1^2 - 0^2 = 0$$

$$I_c = \frac{14}{20}(.4082) + \frac{6}{20}(0) = .2857$$

$$I_G = .5 - .2857 = .2143$$

4.



$$P_0 = \frac{3}{11} \quad P_1 = \frac{8}{11}$$

$$I_L = 1 - \left(\frac{3}{11}\right)^2 - \left(\frac{8}{11}\right)^2 = .3967$$

$$P_0 = \frac{7}{9} \quad P_1 = \frac{2}{9}$$

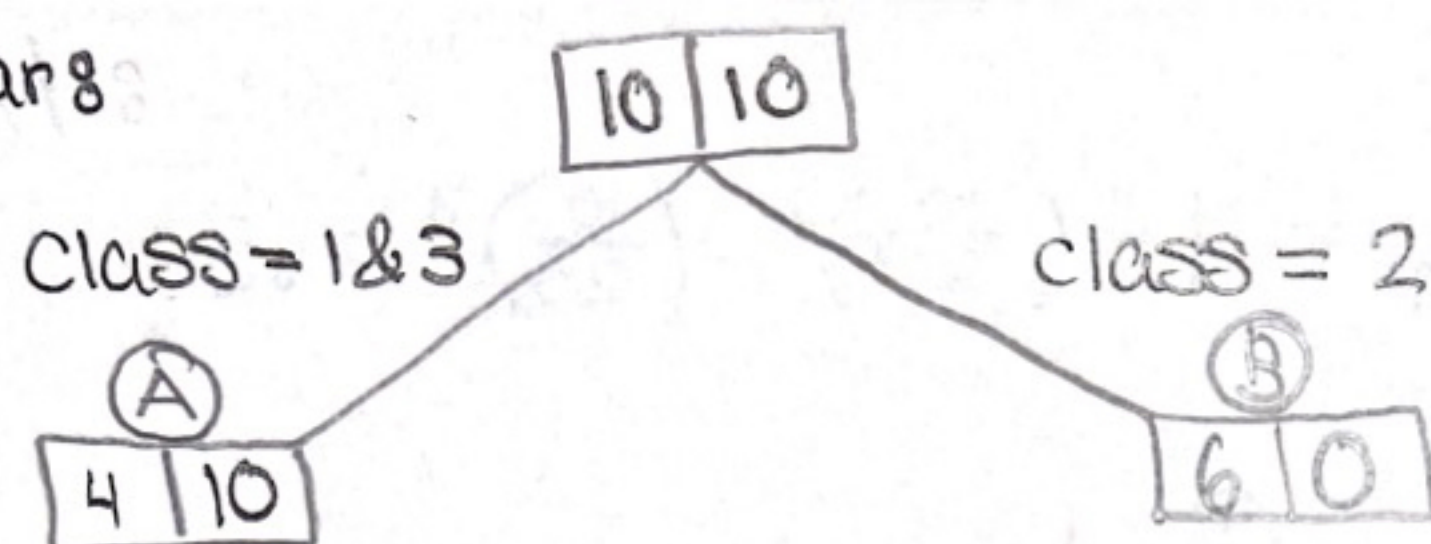
$$I_R = 1 - \left(\frac{7}{9}\right)^2 - \left(\frac{2}{9}\right)^2 = .3457$$

$$I_c = \frac{11}{20}(.3967) + \frac{9}{20}(.3457) = .3738$$

$$I_G = .5 - .3738 = .1262$$

→ option 3 is best split

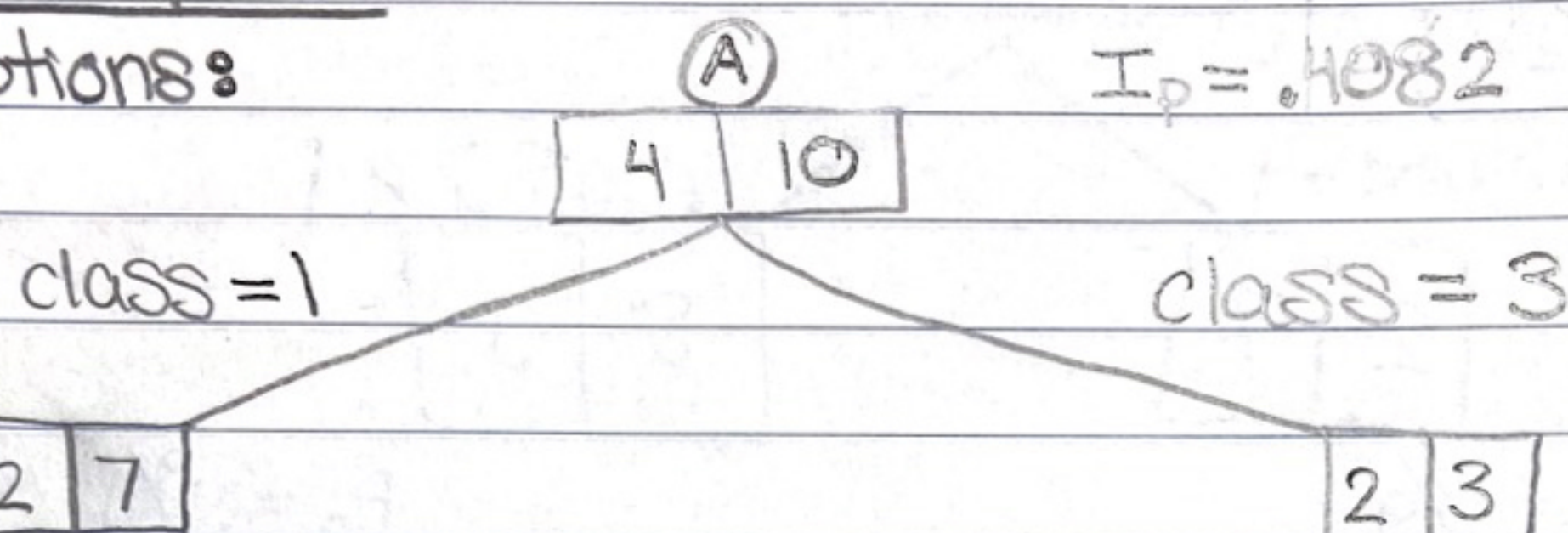
Tree So Far:



Second Split Options:

Split A Options:

A1.



$$I_p = .4082$$

$$P_0 = 2/9 \quad P_1 = 7/9$$

$$I_L = 1 - (2/9)^2 - (7/9)^2 = .3457$$

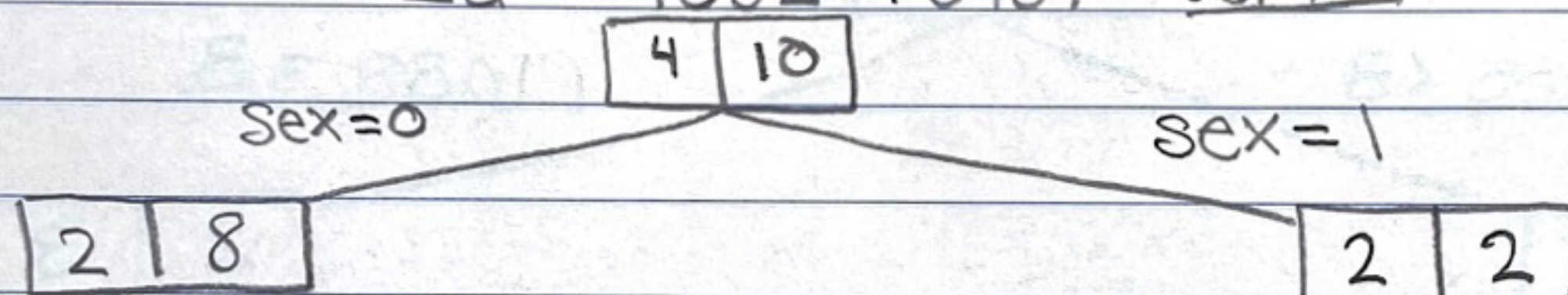
$$P_0 = 2/5 \quad P_1 = 3/5$$

$$I_R = .48$$

$$I_c = 9/14 (.3457) + 5/14 (.48) = .3937$$

$$I_G = .4082 - .3937 = .0145$$

A2.



$$P_0 = 2/10 \quad P_1 = 8/10$$

$$I_L = 1 - (2/10)^2 - (8/10)^2 = .32$$

$$P_0 = 2/4 \quad P_1 = 2/4$$

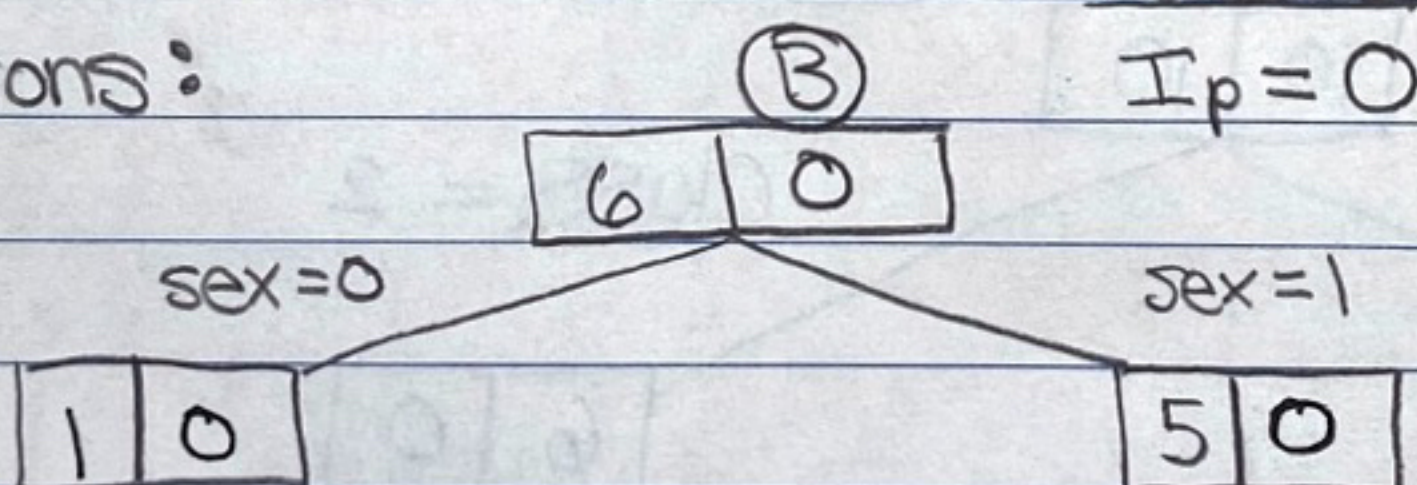
$$I_R = 1 - (1/2)^2 - (1/2)^2 = .5$$

$$I_c = 10/14 (.32) + (4/14) (.5) = .3714$$

$$I_G = .4082 - .3714 = .0368$$

Split B Options:

B1.



$$I_p = 0$$

$$P_0 = 1 \quad P_1 = 0$$

$$I_L = 1 - 1^2 - 0^2 = 0$$

$$P_0 = 1 \quad P_1 = 0$$

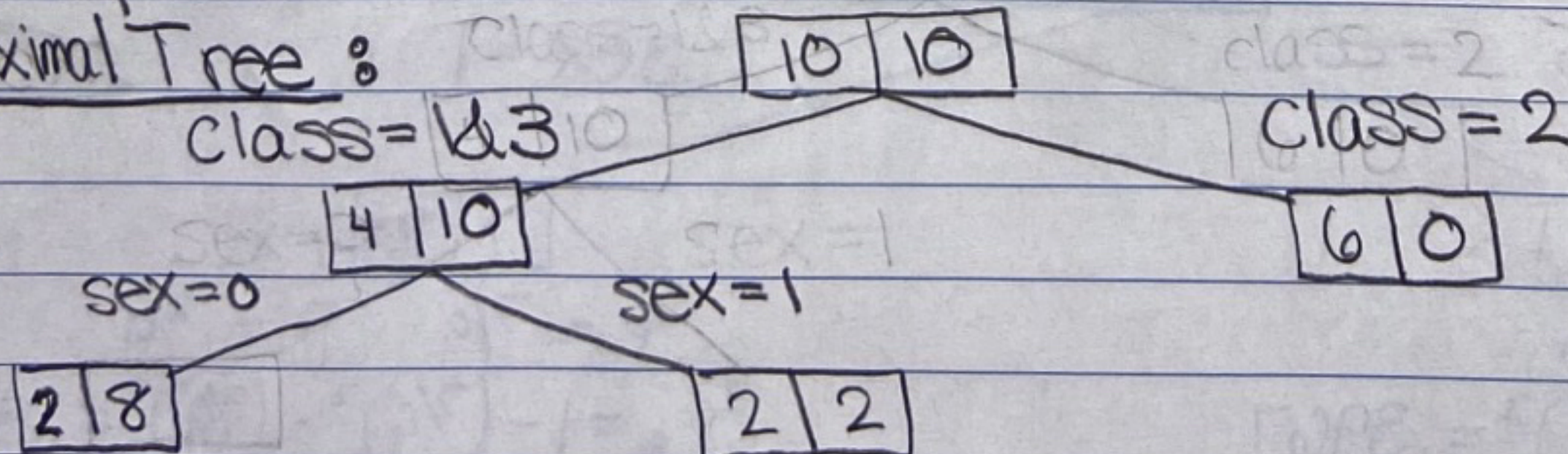
$$I_R = 1 - 1^2 - 0^2 = 0$$

$$I_c = 1/6 (0) + 5/6 (0) = 0$$

$$I_g = 0 - 0 = 0$$

→ Split A @ sex

Maximal Tree:



② misclassification rate = $\frac{2+2+0}{20} = 20\%$

③ Subtrees:

1.

Class = 1 & 3

class = 2

4 | 10

6 | 0

2.

Class = 1 & 3

class = 2

4 | 10

6 | 0

sex = 0

sex = 1

2 | 8

2 | 2

↳ predicts survived

④

Subtree 1:

obs 1: correct

obs 2: correct

obs 3: missed

obs 4: correct

Subtree 2:

obs 1: correct

obs 2: correct

obs 3: missed

obs 4: correct

→ same misclassification → pick simpler tree

→ optimal tree is subtree 1:

class = 1 & 3

class = 2

4 | 10

6 | 0